Consumer Sentiment and Short-Term Market Responses to AI Feature Announcements



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Abstract

Major technology firms have rapidly integrated AI features into consumer products, with concerns emerging that market valuations may exceed fundamental value. The Bank of England's Financial Policy Committee recently warned that equity valuations for AI-focused technology companies "appear stretched." This paper determines the relationship between consumer sentiment and firm performance in the context of artificial intelligence feature announcements. I determine consumer sentiment directed towards brands making these announcements from Reddit, and abnormal returns from the NASDAQ for these brands. Using a Granger Causality and Vector Autoregression framework, I find that consumer sentiment predicts next-day abnormal returns under lag structures that allow for slower adjustment dynamics, while contemporaneous sentiment shows no significant relationship. This study contributes to the literature on behavioral finance and market efficiency and has implications for investors seeking to anticipate short-term price movements and firms managing product launches.

Keywords: Consumer Sentiment, Artificial Intelligence, AI Product Announcements, Abnormal Returns, Granger Causality, Online Attention

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1. Introduction

In recent years, firms have increasingly incorporated artificial intelligence (AI) features into their products in ways that consumers cannot avoid. While some of these additions respond to consumer demand, many appear more like strategic moves to stay competitive, regardless of whether consumers want them. Public reaction often reflects this tension. Some users welcome artificial intelligence features as helpful tools, while others view them with skepticism or outright resistance. This polarized environment makes the consumer response to AI feature announcements especially salient; it reflects not just adoption or rejection of new features, but broader attitudes toward how much technology firms should influence daily life. By evaluating high-profile AI case studies (Microsoft Copilot, Google Gemini, Snapchat My AI, Spotify AI DJ, and Adobe Firefly), I ask whether consumer sentiment around AI helps explain short-term firm performance.

Artificial intelligence refers to machines that simulate human intelligence. Starting in 2020, the release of OpenAI's GPT-3 indicated a turning point wherein large language models (LLMs) trained on vast datasets could generate humanlike text responses. From 2021 through 2024, the number of LLMs increased drastically in what many called an "AI boom." These models began appearing directly in consumer products—Microsoft Office added Copilot, Google integrated Gemini into Search, Adobe launched Firefly for image generation. This rapid integration has sparked debate. While firms tout innovation and efficiency gains, critics highlight costs: massive energy consumption from data centers, environmental impacts, and questions about whether consumers actually benefit from ubiquitous AI.

The scale and speed of AI adoption has led some analysts to question whether the technology represents genuine value creation or speculative excess. Some even suggest that the rapid growth of AI technologies constitutes an "AI bubble." Firms driving this rapid development and innovation are worth an immense portion of the technology sector's earnings; NVIDIA, widely considered the dominant leader in AI computing, is worth just over 4.5 trillion dollars as of December 2025. Just 11 months earlier, in January of 2025, NVIDIA had been worth under 3 trillion dollars (Yahoo Finance). On October 2nd, 2025, the Bank of England's Financial Policy Committee declared that "The risk of a sharp market correction has increased… equity market valuations appear stretched, particularly for technology companies focused on Artificial Intelligence (AI)" (Bank of England 2025). The managing director of the International Monetary Fund, Kristalina Georgieva, added that "Today's valuations are heading toward levels we saw during the bullishness about the internet 25 years ago" (Georgieva 2025). These warnings make examining the link between consumer sentiment and stock returns around AI announcements especially timely, as the relationship can reveal whether markets are rationally pricing new AI features or responding to broader speculative momentum.

AI feature announcements provide a unique setting to examine whether consumer sentiment predicts firm performance. These announcements are highly visible, frequent, and tend to provoke strong reactions, both positive and negative. On one hand, many consumers adopt AI tools quickly and rely on them for convenience; on the other, others push back against their ubiquity, citing risks from privacy concerns to misinformation. This divide creates an ideal scenario in which to study how public perception translates into financial outcomes. This link is not obvious; the Efficient Markets Hypothesis (EMH) implies that stock prices should already reflect all publicly available information, meaning

sentiment should have no predictive power if markets are fully efficient. Behavioral finance, however, suggests that markets may often misprocess or underreact to such news (Fama 1970; Barberis and Thaler 2002). Together, these competing theoretical frameworks yield ambiguous predictions, making it unfeasible to determine a priori whether consumer sentiment should contain unpriced market information and thus requiring direct empirical evaluation.

Recent research has evaluated the manner in which artificial intelligence is shaping consumer perceptions and firm outcomes. Related work has examined how consumers perceive AI chatbots on platforms like Twitter and Reddit, finding both enthusiasm and skepticism. Positive perceptions often center on convenience and engagement, while concerns about job displacement and lack of emotional intelligence contribute to negative sentiment (Graham and Stough 2025). At the firm level, evidence suggests that AI adoption can enhance performance through improved operational efficiency, better marketing insight, and resilience in uncertain environments (Mishra et al. 2022; Ho et al. 2022). Together, the literature suggests that consumer reactions to AI innovations are not uniform, but rather involve a balance between curiosity, trust, and skepticism, which are all factors that may ultimately alter the degree to which AI announcements influence firm market performance.

The existing literature has primarily examined long-run firm outcomes associated with broad AI adoption, often focusing on companies whose core business is already centered on AI technologies. These studies provide valuable insights into structural performance effects but overlook how markets respond in the immediate aftermath of AI-related announcements. This gap is particularly relevant given ongoing debates about AI sector valuations and whether current market enthusiasm reflects fundamental value creation or speculative excess. Understanding short-term market reactions to AI announcements can help distinguish between these possibilities. Moreover, prior work tends to assess AI at the firm level (asking whether a company is "AI-oriented") rather than isolating the impact of specific feature announcements. I work to fill this gap by analyzing firms that are not fundamentally AI companies but that integrate AI features into their existing products. This investigates how incremental and product-specific AI announcements affect short-term firm performance.

A second contribution is the explicit incorporation of consumer sentiment as a predictive mode. While much of the prior literature evaluates firm outcomes directly, without attention to how consumers react, my paper tests whether observable shifts in sentiment on platforms such as Reddit can help explain variation in post-announcement performance. Reddit sentiment is particularly informative in this context. Unlike many other social media platforms, Reddit hosts communities with substantial influence on market outcomes, as demonstrated by the GameStop and AMC trading episodes driven by the r/WallStreetBets community (Lyócsa et al. 2022). This features the role of consumer perception in shaping market outcomes and provides a new angle on the interaction between technological innovation, public reaction, and firm performance. By linking feature-level AI announcements with consumer sentiment and short-term financial responses, my study advances the literature beyond broad categorizations of AI adoption and demonstrates the informational value of consumer reactions in real time.

Past research shows that consumer sentiment can be measured from online text through dictionary-based approaches and machine learning methods. Dictionary methods count positive and negative words

but often struggle with context such as sarcasm (Pang and Lee 2008). Prior research suggests combining multiple sentiment analysis methods to improve reliability (Liu 2015). VADER (Valence Aware Dictionary and Sentiment Reasoner) is specifically designed for social media text and handles informal language, slang, and emoticons effectively (Hutto and Gilbert 2014). TextBlob provides complementary measures of polarity and subjectivity, and its ability to parse linguistic structure helps identify sarcasm and tonal shifts. Several recent studies have applied these methods specifically to Reddit data, showing that user posts can provide reliable measures of consumer attitudes toward brands and products (Timoshenko and Hauser 2019; Wang and Li 2020). Together, this literature provides the foundation for my approach to extracting sentiment from Reddit posts related to AI feature announcements.

Data is collected covering an event window of 4 weeks. Consumer sentiment is extracted from Reddit data collected from the Reddit API. For each of the 5 case studies, approximately 30 search terms are used across 15 subreddits. Daily sentiment scores are determined for each firm, then merged with daily stock metrics. These include the daily stock price, trading volume, and abnormal returns using the NASDAQ as a benchmark.

I first conduct fixed effects regressions to assess whether contemporaneous sentiment predicts abnormal returns, including lagged returns in one specification. Then, I evaluate whether knowing consumer sentiment from Reddit provides more information about a firm's market performance than simply knowing past market performance through Granger causality tests, estimating both one-lag and two-lag specifications. I also test for reverse causality to determine whether returns predict future sentiment, which would complicate causal interpretation. Finally, I estimate a Vector Autoregression (VAR) model to examine the dynamic relationship between sentiment and returns across multiple time periods and to assess robustness to alternative lag structures.

Results show that lagged sentiment significantly predicts abnormal returns in panel Granger causality tests under the two-lag specification, with effects concentrated at a one-day horizon. A one-unit increase in sentiment predicts approximately 2.2 percentage points higher abnormal returns the following day. However, one-lag Granger tests indicate no predictive relationship, suggesting that sentiment effects emerge only when slower adjustment dynamics are permitted. Reverse causality tests consistently indicate that returns do not significantly predict future sentiment, suggesting a primarily unidirectional relationship from sentiment to returns. Fixed effects regressions without lags show no significant contemporaneous relationship, indicating that the predictive power operates through lagged effects rather than same-day reactions. Firm heterogeneity analysis reveals that this relationship varies substantially across companies, with Adobe showing positive sentiment effects while Microsoft, Google, and Snapchat exhibit negative relationships. VAR estimates confirm these patterns qualitatively, though results are somewhat weaker due to reduced sample size from firm averaging.

These results have significant implications for understanding the speed at which financial markets incorporate consumer sentiment into stock prices. The finding that sentiment predicts returns only under longer lag structures suggests that financial markets do not immediately incorporate publicly available consumer reactions, challenging strong-form market efficiency. This short delay indicates a potential means through which investors could anticipate short-term price movements following AI announcements. For firms, these results indicate that consumer reactions to product announcements can

shape short-term market value, though this relationship varies substantially by firm and context. The heterogeneity in sentiment effects suggests that the market impact of consumer reactions depends on firm-specific factors such as product type and user base. More broadly, studying sentiment around AI features offers insight into how consumers process technological changes and whether they embrace or resist innovations that are increasingly embedded into daily life.

The remainder of this paper proceeds as follows. Section 2 describes the data sources and the process of collecting Reddit posts and financial metrics. Section 3 outlines the methods used to quantify sentiment and evaluate its predictive relationship with firm performance. Section 4 presents the results of the analysis. Section 5 discusses the implications of the findings and concludes.

2. Data

I focus on five major AI feature announcements from 2023: Microsoft 365 Copilot (March 16), Spotify AI DJ (February 22), Google Gemini (December 6), Snapchat My AI (February 27th), and Adobe Firefly (March 21). Each represents a visible consumer-facing AI launch that received substantial public attention. The goal was to select cases where the announcement itself (not surrounding events) drove both consumer discussion and market response.

The use of 2023 as a time frame is particularly compelling for analysis due to the context of AI presence, as many consumers were able to first interact directly with AI. ChatGPT, an Open-Source AI Chatbot, was released in November of 2022, and quickly gained popularity; it had 100 million monthly active users by January of 2023. At the time, ChatGPT was the fastest growing consumer app ever created (Hu 2023). The virality of ChatGPT was widely discussed as the public debated whether the usefulness of the tool exceeded its negative externalities (e.g., the intensive computation required via data centers to accommodate the high amount of traffic received by the site). As such, the increase in public awareness of AI and its associated cost-benefit analysis, as well as the increase in AI feature embedding within existing firms, provides an appealing environment in which to determine the relationship between consumer sentiment and firm financial outcomes following AI feature announcements.

Each announcement selected met three criteria. First, none coincided with confounding events such as earnings releases or leadership changes. For example, Microsoft and Spotify's Q1 2023 earnings occurred more than a month after their respective announcements, while Google's Q4 2023 earnings followed in early 2024. Second, each firm is publicly traded on major U.S. exchanges, which ensures reliable market data. Third, all have active online communities that discuss new products, which allows for sufficient measurement of a consumer sentiment metric.

The five firms represent different segments of the technology sector. Microsoft's Copilot integrates AI into productivity tools, Spotify's AI DJ applies it to entertainment and personalization, Google's Gemini introduces a general-purpose language model, Adobe Firefly brings AI to creative software, and Snapchat My AI embeds AI into social media platforms. This variation in sector provides breadth in examining how consumer sentiment responds to AI features across different product categories and user bases. It also allows for more nuance in the evaluation of firm heterogeneity and the differences in sector response to consumer sentiment.

Announcements were spaced across the year, with some overlap between event windows. While complete separation would be ideal, the firm-level variation and fixed-effects approach allow for isolation of company-specific sentiment effects. The staggered timing still provides sufficient independence to identify how individual firms' announcements affect their own stock performance, as each event captures distinct consumer reactions to different products in different market contexts.

2.2 Data Collection

2.2.1 Consumer Sentiment

Posts and comments were collected from Reddit using the Python Reddit API Wrapper (PRAW). Data were gathered over a four-week window surrounding each announcement, two weeks before and two weeks after. This window allowed for collection of a sufficient number of observations but was still limited enough to minimize the risk of including confounding events.

Approximately 30 "congruent" search terms were applied across 15 subreddits (Reddit communities built for discussing specific topics) per case, targeting relevant technology and brand-specific communities. For example, each term-specific search included the isolated name of the firm, the name of the term followed by 'AI', the name of the firm followed by the feature name, and more. These were kept congruent between cases, ensuring balanced results from data collection. Targeted subreddits included, but were not limited to, r/technology, r/news, r/tech, and r/ArtificialInteligence [sic]. Filtering steps removed duplicates and irrelevant posts, including those that mostly focused on other firms. Competitive indicators (e.g. "anthropic") would disqualify posts with significant mention of outside firms and products beyond mere reference. This allows for posts including comparisons between features and models but excludes posts primarily discussing irrelevant firms or features.

Comments from posts with high-engagement threads (50+ net upvotes) were retained to ensure visibility. Across the five cases, this process yielded 659 posts and approximately 9,000 comments, totaling 9,705 pieces of content for analysis. Table 1 presents the sample composition and data collection summary for each firm, showing substantial variation in discussion volume across announcements.

Table 1: Sample Composition and Data Collection Summary

Firm	Feature	Event Date	Reddit Posts	Reddit Comments	Total Content	Trading Days
Microsoft	365 Copilot	2023-03-16	203	3,390	3,593	20
Spotify	AI DJ	2023-02-22	61	591	652	19
Google	Gemini	2023-12-06	262	4,189	4,451	19
Snapchat	My AI	2023-02-27	51	202	253	19
Adobe	Firefly	2023-03-21	82	674	756	20

Google's Gemini generated the most discussion with 4,451 total pieces of content, followed by Microsoft's Copilot with 3,593. Adobe, Spotify, and Snapchat generated considerably less activity, with 756, 652, and 253 pieces of content respectively. These differences likely reflect each firm's user base size, the prominence of the announcement, and the perceived significance of the AI feature to existing product ecosystems.

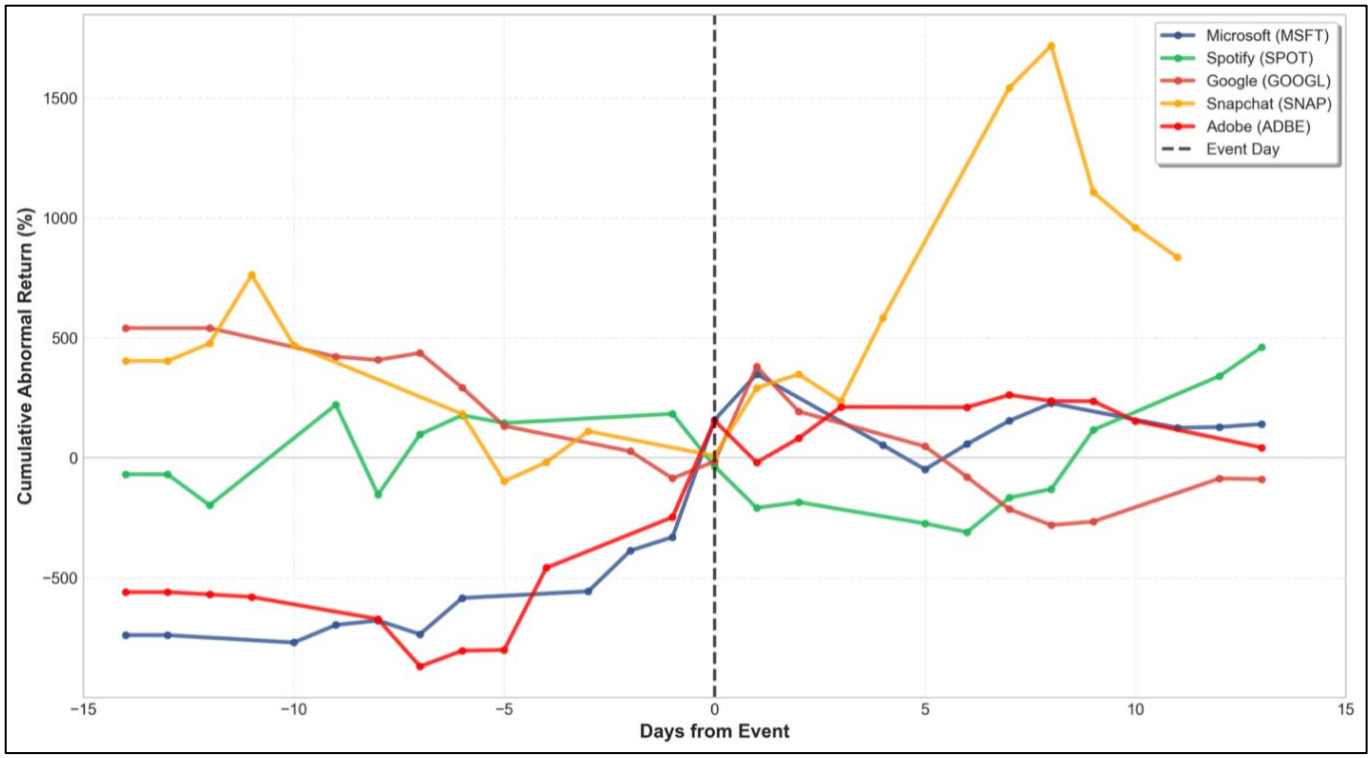
2.2.2 Firm Performance

Firm performance data were obtained from Yahoo Finance through the yfinance Python library. For each company, daily stock prices, trading volumes, and market benchmarks were collected for the same four-week event window.

Performance is measured through abnormal returns, calculated as the firm's daily return minus the NASDAQ return for that day. The NASDAQ serves as an appropriate benchmark, as technology accounts for over half of total market weight and "tech giants" such as Apple, Microsoft, and NVIDIA contribute (Nasdaq, Inc. 2025). The use of abnormal returns isolates firm-specific movement from overall market trends. Cumulative abnormal returns (CARs) were then computed by summing daily abnormal returns across the event window to assess whether initial reactions persisted.

Figure 1 presents the cumulative abnormal returns for each firm throughout the event window, with day 0 marking the announcement date.

Figure 1: Cumulative Abnormal Returns Around AI Announcement



The trajectories reveal substantial heterogeneity in market responses. Snapchat exhibits the most dramatic pattern, with CARs rising sharply after the announcement to peak around day 8 at approximately 17%, before declining in the final week. Microsoft and Adobe show more modest positive cumulative returns by the end of the window, while Spotify's trajectory is relatively flat, hovering near zero. Google's CARs decline in the post-announcement period, ending slightly negative. This variation across firms motivates the panel analysis approach used in subsequent sections, which exploits both time-series and cross-sectional differences to identify sentiment effects.

Stock data were adjusted for splits and dividends, and non-trading days were excluded. When Reddit activity occurred on weekends or holidays, sentiment from those dates was matched to the nearest trading day.

2.3 Sentiment Analysis

Two methods were used to score sentiment: VADER (Valence Aware Dictionary and Sentiment Reasoner) and TextBlob. VADER is optimized for social-media language and captures informal expressions, while TextBlob provides polarity and subjectivity scores based on linguistic patterns.

For each Reddit post and comment, both tools generated sentiment scores ranging from −1 (negative) to +1 (positive). Their averages formed a single "consensus" score. Daily average sentiment was then calculated by date, creating a time series that could be merged with stock data for analysis.

2.4 Descriptive Patterns

Table 2 presents descriptive statistics for the key variables used in the analysis.

Table 2: Descriptive Statistics

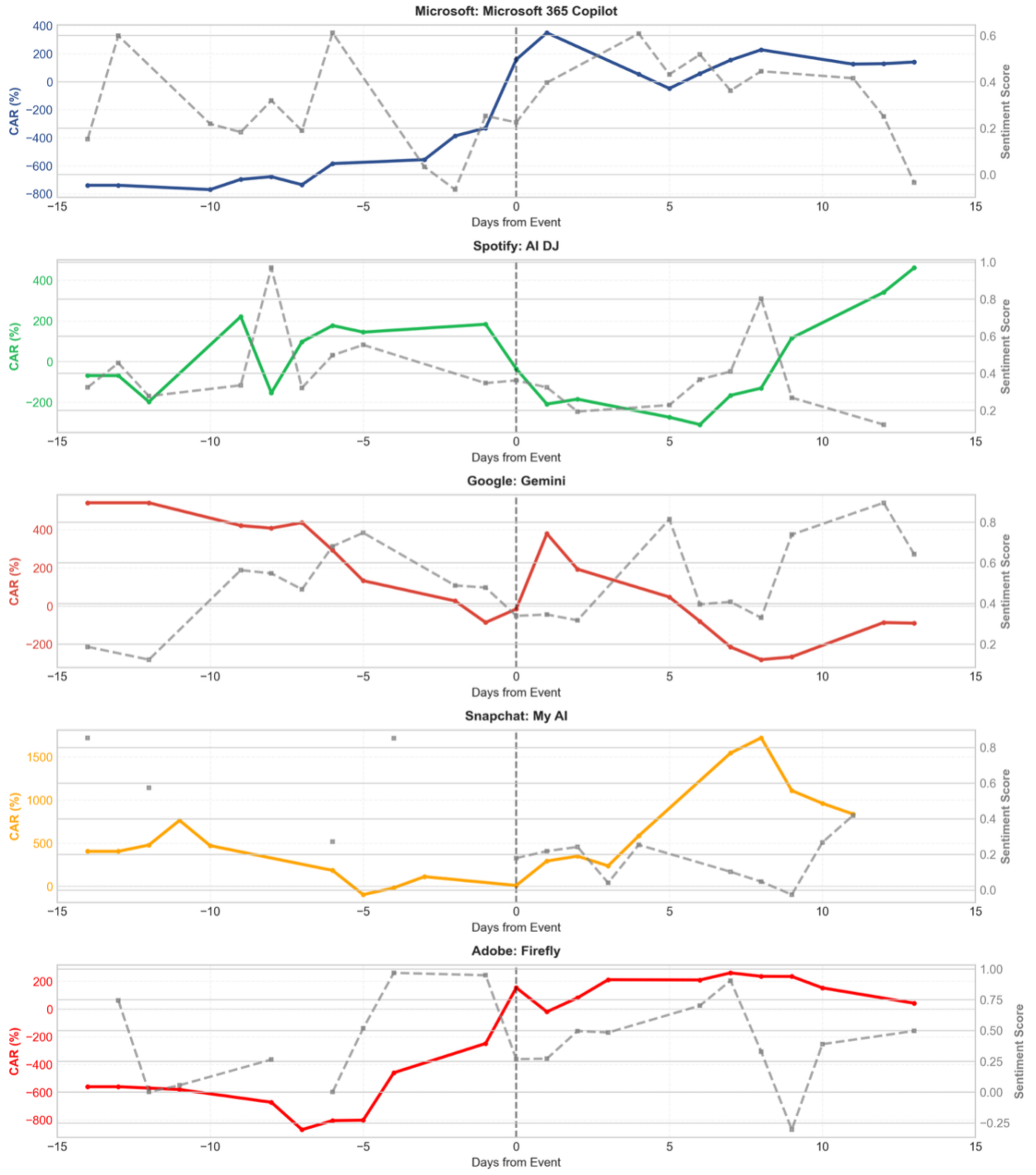
Variable	N	Mean	Median	Std Dev	Min	Max
A: Returns						
Daily Return	92	0.0023	0.0005	0.0248	-0.0572	0.0948
Abnormal Return	92	0.0017	0.0006	0.0197	-0.0611	0.0960
Market Return	92	0.0006	0.0019	0.0118	-0.0250	0.0248
CAR	97	0.3206	0.5703	4.7138	-8.7034	17.1819
B: Trading Activity						
Volume (Millions)	97	20.2258	22.5019	17.7869	0.8700	81.5240
Days from Event	97	-0.66	0.00	8.16	-14.00	13.00
C: Sentiment						
VADER Score	254	0.3820	0.3218	0.3433	-0.9988	0.9999
TextBlob Polarity Score	254	0.1153	0.1016	0.1075	-0.3000	0.7936
Daily Post Count	254	30.2087	7.000	79.2973	1.000	511.0000

Abnormal returns average 0.17% per day with substantial variation (SD = 1.97%), while cumulative abnormal returns range from -8.70% to 17.18% across the event windows. Daily sentiment scores from VADER average 0.38 (on a -1 to +1 scale), indicating generally positive consumer reactions to AI announcements, though with considerable heterogeneity (SD = 0.34). TextBlob polarity scores show similar patterns, averaging 0.12 with a range from -0.30 to 0.79. Trading volume averages approximately 20.2 million shares per day across the five firms, reflecting active market participation during these event windows.

Sentiment patterns varied across the five announcements, reflecting differences in product context and user base reactions. Adobe Firefly generated highly positive sentiment, with consumers viewing it as a valuable creative tool. Spotify AI DJ also received predominantly positive reactions. Microsoft 365 Copilot elicited more mixed sentiment, with enthusiasm for productivity gains tempered by concerns about workflow disruption. Google Gemini faced notable skepticism, likely due to intense competition in the large language model space and questions about differentiation from existing products. Snapchat My AI showed the most volatile sentiment patterns, with reactions shifting considerably throughout the event window.

Figure 2 displays the time-series relationship between cumulative abnormal returns and daily sentiment for each firm.

Figure 2: Cumulative Abnormal Returns and Sentiment Around AI Announcements



The panels convey substantial heterogeneity in how sentiment and returns evolved around the announcements. Microsoft shows a sharp positive CAR spike at announcement followed by relatively stable sentiment that declines late in the window. Spotify exhibits rising CARs in the post-announcement period alongside sustained positive sentiment. Google's CARs peak at announcement but decline afterward, while sentiment remains volatile. Snapchat's dramatic CAR increase in the second week corresponds with initially low but rising sentiment. Adobe shows steady positive CARs with highly positive and stable sentiment throughout. These divergent patterns suggest that the relationship between sentiment and returns varies by firm and may depend on announcement timing, product reception, and market context.

Stock patterns varied widely. Microsoft's share price rose sharply on the announcement day but flattened afterward, suggesting short-term enthusiasm. Spotify's returns were slightly negative throughout the period despite overwhelmingly positive sentiment, implying that investors remained cautious about revenue impacts. Google's returns dipped at announcement but later recovered, ending the period modestly positive.

Comparing raw and market-adjusted returns (see Figure 1) shows that Microsoft's post-announcement gains were largely firm-specific, while Spotify underperformed the market and Google's movements gradually aligned with it.

2.5 Variable Justification

The dependent variable, firm performance, is measured through abnormal returns. This metric isolates changes in firm value from broader market shifts and captures the short-term effects of news events. Trading volume serves as a secondary measure to identify cases where announcements generate attention but limited price movement.

The independent variable, consumer sentiment, measures the average daily tone of Reddit discussions. Measuring sentiment in this way connects the public's immediate reaction to market outcomes and tests whether observable consumer discourse contains predictive information about firm value. Naturally, Reddit sentiment may not be the most representative metric of overall consumer opinion, as sourcing data from technology and firm specific subreddits may skew the data toward more technologically aware or passionate individuals.

3. Empirical Strategy

This section outlines the empirical strategy used to test whether consumer sentiment surrounding AI feature announcements predicts short-term firm performance.

The analysis proceeds in three stages: (1) a baseline fixed-effects regression examining within-firm associations between sentiment and performance; (2) Granger causality tests assessing temporal precedence; and (3) a vector autoregression (VAR) model included as a robustness check to capture dynamic relationships and evaluate impulse response.

3.1 Baseline Regression

The empirical analysis begins by estimating whether daily changes in consumer sentiment surrounding AI feature announcements predict short-term firm performance. Two specifications are estimated. The first examines the contemporaneous relationship:

$$\text{Perf}_{it} = \alpha + \beta_1 \text{Sentiment}_{it} + \mu_i + \epsilon_{it}$$

The second specification includes lagged abnormal returns to control for momentum effects:

$$\text{Perf}_{it} = \alpha + \beta_1 \text{Sentiment}_{it} + \beta_2 \text{Perf}_{it-1} + \mu_i + \epsilon_{it}$$

where Perf_{it} is a daily measure of firm performance (abnormal returns, trading volume, or stock price change) for firm *i* on day *t*, and Sentiment_{it} represents the average Reddit sentiment toward that firm on the same day, computed using the VADER and TextBlob sentiment analyzers. The lagged performance variable Perf_{it-1} captures momentum in returns. μ_i captures firm fixed effects that control for unobserved, time-invariant firm traits such as brand reputation, and ϵ_{it} is an idiosyncratic error term.

Because the sample consists of five firms observed over short event windows, time fixed effects are excluded; including them would absorb most of the day-to-day variation that identifies the relationship of interest. The main identifying assumption is that, conditional on fixed effects and lagged financial controls, changes in sentiment are exogenous to unobserved shocks that simultaneously affect firm performance (the "exogeneity assumption"). The model also assumes stationarity of both sentiment and performance within each event window. In essence, it is presupposed that the statistical properties of the data series (mean, variance, etc.) do not change over time. However, simultaneity remains a potential concern: firm performance may itself influence consumer sentiment. This motivates the use of Granger causality tests in the next section, which explicitly examine whether sentiment leads performance rather than merely responding to it. Standard errors are clustered at the firm level to account for serial correlation over time.

3.2 Granger Causality

To examine whether consumer sentiment changes precede or simply reflect firm performance, I estimate a series of Granger causality tests. These models assess whether lagged sentiment values improve the prediction of current firm performance beyond what is explained by the firm's own history. The first equation is specified as

$$\text{Perf}_{it} = \alpha_0 + \sum_{k=1}^{p} \beta_k \text{Perf}_{t-k} + \sum_{k=1}^{p} \lambda_k \text{Sentiment}_{t-k} + \epsilon_t$$

where p denotes the number of lags. The first summation term represents the autoregressive portion of the model, determining how much past firm performance informs current performance. The second summation term tests whether lagged sentiment values contain additional predictive power beyond past sentiment; rejecting the joint null hypothesis that all $\lambda_k = 0$ indicates that sentiment adds predictive power, meaning sentiment Granger-causes performance.

A complementary equation serves as a robustness check for the model by reversing the direction of dependence to test whether past firm performance helps explain future sentiment:

$$\text{Sentiment}_t = \delta_0 + \sum_{k=1}^{p} \phi_k \text{Sentiment}_{t-k} + \sum_{k=1}^{p} \theta_k \text{Perf}_{t-k} + \nu_t$$

If coefficients θ_k are jointly insignificant, it suggests that the causal flow runs primarily from sentiment to performance rather than the reverse.

Together, the two specifications directly address the simultaneity concern present in the regression proposed in Section 3.1. This framework allows for a more credible analysis of whether shifts in online sentiment anticipate subsequent stock responses or react to them. Evidence that sentiment Granger-causes performance would imply that consumer reactions contain forward-looking information that markets do not immediately price in.

3.2.1 Granger Causality Lag Length Selection

A necessary consideration in implementing panel Granger causality tests in the number of lags. Too few lags risk omitting relevant dynamics, while too many lags reduce the degrees of freedom and introduce noise. Following standard time-series practice, I use the Akaike Information Criterion (AIC) for lag selection. The AIC estimates the expected prediction error of the model, measuring how well the model would perform on unseen data. It balances two competing concerns: model fit (how well the model explains current data) and model complexity (how many parameters are estimated). An increase in lags improves fit to current data but may lead to overfitting as the model captures noise rather than actual patterns. The AIC penalizes this complexity by adding a term proportional to the number of parameters. The lag length that minimizes the AIC achieves the optimal tradeoff. Lower AIC values indicate better expected out-of-sample performance. The AIC is calculated using:

$$\text{AIC} = -2\ln(L) + 2k$$

where ln(L) is the log-likelihood and k is the number of parameters. The minimum AIC will locate the model that maximizes likelihood while avoiding excess parameters.

Notably, Granger causality studies in applied macroeconomics and finance routinely estimate multiple lag structures, even when information criteria favor a more parsimonious choice, because causality inference is sensitive to lag length (Lütkepohl 2005; Toda and Yamamoto 1995; Rapach and Zhou 2013). It is standard practice to test robustness across different lag lengths, as the AIC-optimal specification may not fully capture slower adjustment processes in economic data (Kilian and Lütkepohl 2017; Hamilton 1994). This follows from the fact that Granger causality tests may underestimate dynamic relationships when restricted to too few lags.

Table 3 presents the AIC and BIC (Bayesian Information Criterion) for 1-2 lag specifications. BIC, an alternate specification that tends to select more parsimonious models due to a stronger complexity penalty, is included for completeness. Testing is limited to 1-2 lags to avoid overfitting.

Table 3: Granger Causality Lag Lengths

Lags	Log-Likehood	AIC	BIC	Observations
1	176.76	-347.51*	-340.99*	65
2	121.59	-233.17	-22.436	43

Asterisks (*) denote the optimal value for each criterion.

Both the AIC and BIC favor the 1 lag specification, with respective minimums of -347.51 and -340.99. However, given the precedent that Granger causality inference is sensitive to lag length (Lütkepohl 2005; Toda and Yamamoto 1995; Rapach and Zhou 2013), I estimate both the 1-lag specification and the more conventional 2-lag specification used in prior literature. This avoids relying on a single modeling choice, allowing me to assess whether the predictive relationship between sentiment and returns is robust across reasonable lag structures.

3.3 Vector Autoregression

As a robustness check, I estimate a Vector Autoregression (VAR) model to capture the full dynamic relationship between sentiment and returns. While the Granger causality tests provide evidence on directional precedence, the VAR framework allows both variables to influence each other over multiple time periods and provides a more flexible specification for modeling their joint evolution.

To implement the VAR, I first collapse the panel data by averaging across firms for each day in the event window, creating a single time series of approximately 41 observations per variable. This averaging is requisite as standard VAR models require a single time series for each variable rather than previously used panel data. The optimal lag length is selected using the Akaike Information Criterion (AIC). Based on the AIC, I estimate the VAR with 1 lag.

The VAR framework enables two complementary analyses. First, Granger causality tests within the VAR formally assess whether lagged values of sentiment help predict current returns (and vice versa), controlling for the system's full dynamic structure. These tests use chi-squared statistics to evaluate joint significance of lagged coefficients. Second, orthogonalized impulse response functions (IRFs) trace out how abnormal returns respond dynamically to an unexpected one-standard-deviation shock in sentiment. The IRF shows the trajectory of this response over the following 10 days, with 95% confidence intervals constructed via bootstrapping.

The VAR approach has both advantages and limitations relative to the panel Granger tests. It captures more detailed dynamics by allowing feedback between variables. However, collapsing across firms reduces both sample size and statistical power, and the averaging may obscure firm-specific heterogeneity that the panel models exploit. The VAR results should therefore be interpreted as complementary evidence.

3.3.1 Vector Autoregression Lag Length Selection

The VAR model requires a different data structure than the panel Granger tests. After removing firm fixed effects by demeaning within firms and aggregating to a single time series per day, the sample size decreases substantially. Because information criteria depend on both sample size and parameter count, lag selection must be recomputed.

Consistent with standard VAR methodology (Lütkepohl 2005; Kilian and Lütkepohl 2017), I compute the AIC and BIC for VAR models with 1 and 2 lags (Table 4).

Table 4: Vector Autoregression Lag Lengths

Lags	Log-Likehood	AIC	BIC	Observations
1	134.93	-257.85*	-247.57*	41
2	97.69	-175.37	-161.70	29

Asterisks (*) denote the optimal value for each criterion.

Table 4 shows that both criteria select a 1-lag VAR. This choice is consistent with the relatively short time series, as high-lag specifications would consume degrees of freedom quickly.

For comparability with the Granger Causality tests, and as prior work frequently uses VAR(1) models to study short term shock transmission (e.g., Sims 1980; Stock and Watson 2001), I implement a VAR(1) specification for impulse response analysis. As in section 3.2.1, the results are later contrasted with a VAR(2) model as a robustness check.

4. Results

To examine whether consumer sentiment surrounding AI feature announcements predicts short-term firm performance, I first estimate a baseline fixed-effects model and then proceed to formal tests of temporal precedence through Granger causality analysis.

4.1 Baseline Relationship

Table 5 presents results from the fixed-effects regression examining the contemporaneous relationship between daily sentiment and abnormal returns. Column (1) shows the baseline specification, while Column (2) includes lagged abnormal returns to account for a momentum effect. The previous day's return likely affects the current day's return; if sentiment is picking up this momentum, the lagged abnormal returns absorb it.

Table 5: Baseline Fixed Effects Regression

	(1) **Without Lag**	(2) **With Momentum Control**
Daily Average Sentiment	-0.0032	0.0080
	(0.0130)	(0.0088)
Lagged Abnormal Returns		0.0723
		(0.0570)
Constant	0.0035	0.0007
	(0.0049)	(0.0032)
Firm Fixed Effects	Yes	Yes
Observations	88	65
Within R-squared	0.0012	0.0171
Number of Firms	5	5

Standard errors in parentheses
Dependent variable: Daily abnormal return
Robust standard errors clustered at firm level in parentheses
* p<0.10, ** p<0.05, *** p<0.01

The baseline results indicate that there is not a significant contemporaneous relationship between sentiment and abnormal returns. The coefficient on sentiment is small and statistically insignificant in both specifications ($\beta = -0.003$, $p = 0.819$ without controls; $\beta = 0.008$, $p = 0.415$ with lagged returns). It is implied that same-day sentiment does not directly predict same-day stock performance.

When lagged abnormal returns are included (Column 2), there is weak evidence of momentum ($\beta = 0.072$, $p = 0.273$), as this effect is not statistically significant. The within R-squared is low in both specifications (0.0012 and 0.0171), indicating that these models explain limited variation in daily returns.

4.2 Granger Causality

Because the contemporaneous relationship provides little evidence of association, the natural next step is to determine whether sentiment leads returns rather than coinciding with them. This section evaluates whether sentiment Granger-causes returns using alternate lag specifications.

4.2.1 One Lag Specification

The AIC conducted in section 3.2.1 suggests that likelihood is maximized at a lag length of 1. Table 6 presents results from the Granger causality framework, determining whether sentiment values lagged by one day help predict current returns beyond what past returns alone would suggest.

Table 6: Granger Causality, One Lag Specification

	(1) **Abnormal Return**
L.Daily Average Sentiment	0.009
	(0.006)
L.Daily Abnormal Return	-0.026
	(0.054)
Constant	-0.000
	(0.002)
Firm Fixed Effects	Yes
Number of Lags	1
F-statistic (Test)	2.01
p-value (Test)	0.2291
Significance	Not Significant
Observations	65
Within R-squared	0.014
Number of Firms	5

Standard errors in parentheses
Note: This table tests whether lagged sentiment values Granger-cause abnormal returns. The null hypothesis is that the lagged sentiment coefficient equals zero. F-statistic and *p*-value test the significance of L.Sentiment. Rejection of the null indicates that sentiment contains predictive information about future returns beyond what is captured by past returns alone. Robust standard errors clustered at firm level in parentheses.
* $p<0.10$, ** $p<0.05$, *** $p<0.01$

The one lag specification provides no evidence that sentiment predicts abnormal returns. The coefficient on lagged daily sentiment (0.009) is small and statistically insignificant. This implies that a one-standard-unit increase in the previous day's average Reddit sentiment is associated with only a 0.9-basis-point change in the current day's abnormal return, an effect that is economically and statistically negligible.

Lagged abnormal returns are also insignificant, suggesting little persistence or momentum in returns at the daily level within this sample. The joint F-Test evaluating whether lagged sentiment variables collectively Granger-cause returns yields an F-statistic of 2.01 and a p-value of 0.2291, meaning the null hypothesis (sentiment does not help predict returns) is not rejected at conventional significant levels.

Overarchingly, the above results indicate that when using a single lag, sentiment does not display predictive power for next-day abnormal returns.

4.2.2 Two Lag Specification

Following conventions in the VAR and Granger causality literature, I estimate both the one-lag model selected by AIC and a two-lag model commonly used in empirical applications (Sims 1980; Stock and Watson 2001). In settings where information diffusion or behavioral responses may occur with short

delays, past research frequently incorporates additional lags to avoid prematurely concluding that no causal relationship exists under an overly restrictive specification (Granger 1969; Lütkepohl 2005). Table 7 reports results from the two-lag model.

Table 7: Granger Causality, Two Lag Specification

	(1) **Abnormal Return**
L.Daily Average Sentiment	0.0216**
	(0.0064)
L2.Daily Average Sentiment	-0.0176
	(0.0146)
L.Daily Abnormal Return	-0.0007
	(0.0819)
L2.Daily Abnormal Return	-0.1700
	(0.1520)
Constant	0.0006
	(0.0059)
Firm Fixed Effects	Yes
Lags	2
F-stat (Joint)	7.06
p-value (Joint)	0.0487
Observations	43
Within R-squared	0.1766
Number of Firms	5

Standard errors in parentheses
Dependent variable: Daily abnormal return
H_0: Sentiment does not Granger-cause returns (both lagged sentiment coefficients = 0)
Robust standard errors clustered at firm level in parentheses
* p<0.10, ** p<0.05, *** p<0.01

The results suggest that sentiment Granger-causes abnormal returns when two lags are included. The first lag of sentiment shows a positive and statistically significant effect ($\beta = 0.022$, $p = 0.028$), indicating that a one-standard-deviation increase in sentiment is associated with a 2.2 percentage point increase in abnormal returns the following day. The second lag is negative but not significant ($\beta = -0.018$, $p = 0.295$), suggesting that only the most recent sentiment shift conveys predictive information.

Neither lagged abnormal returns variable is significant, which indicates that past returns alone do not explain next-day performance within this specification. The within-firm R-squared (0.177) is modest but materially larger than in the one-lag specification. This is consistent with the idea that sentiment effects unfold over more than a single day.

The joint test of both lagged sentiment coefficients is statistically significant ($F(2,4) = 7.06$, $p = 0.0487$). The rejection of this null indicates that that sentiment contains predictive power for future returns beyond what can be explained by past returns or firm-level fixed effects.

4.3 Reverse Causality

A reasonable concern is whether the observed relationship reflects reverse causality (that is, whether strong stock performance drives subsequent positive sentiment rather than the opposite). As with the Granger causality presented in section 4.2, I estimate the reverse causality model with both 1 and 2 lags.

4.3.1 One Lag Specification

Table 8 addresses this bidirectionality concern by testing whether lagged abnormal returns help predict current sentiment beyond only utilizing prior sentiment.

Table 8: Reverse Causality, One Lag Specification

	(1) **Daily Sentiment**
L.Daily Abnormal Return	-1.830
	(1.594)
L.Daily Average Sentiment	0.092
	(0.065)
Constant	0.328***
	(0.025)
Firm Fixed Effects	Yes
Number of Lags	1
F-statistic (Test)	1.32
p-value (Test)	0.3148
Significance	Not Significant
Observations	63
Within R-squared	0.036
Number of Firms	5

Standard errors in parentheses
Note: This table tests the reverse causal relationship: whether lagged returns Granger-cause sentiment. The null hypothesis is that the lagged return coefficient equals zero. Finding that returns do not predict sentiment (failure to reject null) combined with evidence that sentiment predicts returns (Table 2) would support unidirectional causality from sentiment to returns. Robust standard errors clustered at firm level in parentheses.
* $p<0.10$, ** $p<0.05$, *** $p<0.01$

The results provide little evidence of reverse causality. The coefficient on lagged abnormal is negative and statistically insignificant ($\beta = -1.830$, $p = 0.315$), indicating that yesterday's returns do not meaningfully influence today's sentiment. The coefficient on lagged sentiment is also statistically

insignificant (β = 0.092, p = 0.332), suggesting that sentiment evolves independently of recent return movements within this specification.

The joint F-test provides supporting evidence of this; the null hypothesis that lagged returns Granger-cause sentiment fails to be rejected (F = 1.32, p = 0.315). The low within-firm R-squared (0.036) further indicates that past returns explain very little of the variation in sentiment. Together, these results suggest that return movements do not systematically drive subsequent consumer discussions on Reddit, reducing concerns that the predictive relationship identified in Section 4.2 is driven by reverse causation.

4.3.2 Two Lag Specification

To assess whether reverse causality occurs when allowing sentiment to respond with a longer delay, I re-estimate the reverse causality model using a two-lag specification. This approach tests whether abnormal returns from one or two days prior help predict current sentiment after accounting for prior sentiment dynamics. Table 9 reports the results from this expanded model.

Table 9: Reverse Causality, Two Lag Specification

	(1) **Daily Sentiment**
L.Daily Abnormal Return	0.8091
	(1.5521)
L2.Daily Abnormal Return	-2.4347**
	(0.8761)
L.Daily Average Sentiment	0.0380
	(0.1799)
L2.Daily Average Sentiment	-0.4660
	(0.2300)
Constant	0.5010***
	(0.0949)
Firm Fixed Effects	Yes
Lags	2
F-stat (Joint)	4.09
p-value (Joint)	0.1079
Observations	43
Within R-squared	0.1882
Number of Firms	5

Standard errors in parentheses
Dependent variable: Daily sentiment (VADER)
H_0: Returns do not Granger-cause sentiment (both lagged return coefficients = 0)
Robust standard errors clustered at firm level in parentheses
* p<0.10, ** p<0.05, *** p<0.01

The reverse causality test provides no strong evidence that returns predict future sentiment. While the second lag of returns shows a marginally significant negative coefficient ($\beta = -2.43$, $p = 0.050$), the joint test fails to reach statistical significance ($F(2,4) = 4.09$, $p = 0.108$). The Granger asymmetry (sentiment predicts returns but returns do not clearly predict sentiment) bolsters the interpretation that consumer reactions contain forward-looking information that markets gradually incorporate.

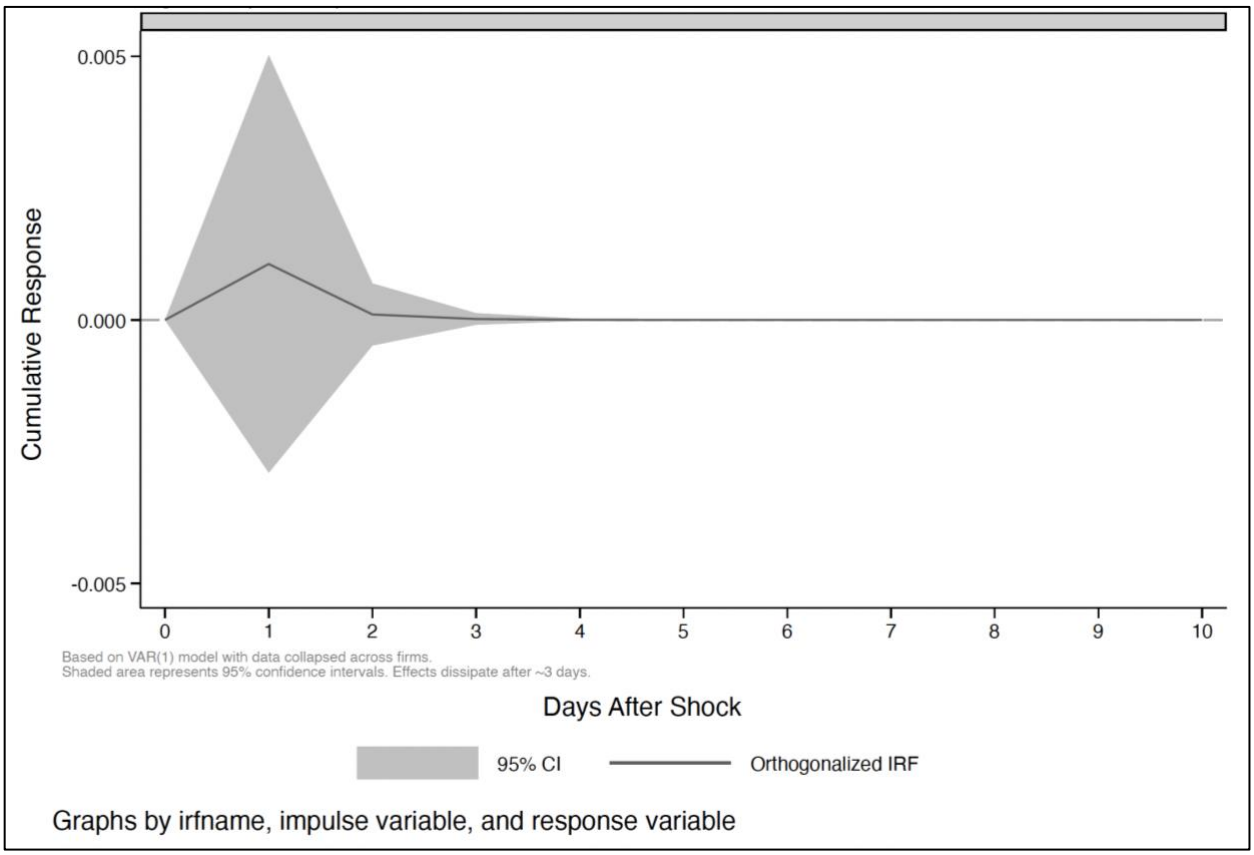
4.4 Vector Autoregression

To capture the full dynamic relationship between sentiment and returns, I estimate a Vector Autoregression (VAR) model that allows both variables to influence each other over time. This is a natural extension of the Granger causality test.

4.4.1 Impulse Response Analysis

Figure 3 presents the impulse response function expressing how abnormal returns respond to a positive shock in sentiment. This IRF originates from a VAR(1) model, as guided by the AIC.

Figure 3: Impulse Response of Abnormal Returns to a Shock in Sentiment



The IRF indicates that a one-standard-deviation increase in Reddit sentiment (representing unusually positive consumer reaction to an AI announcement) generates a positive response in abnormal returns that peaks around day 1 at approximately 0.2 percentage points. The effects then dissipate rapidly, returning to zero by day 3. This short-lived effect is consistent with efficient markets quickly incorporating sentiment signals into prices.

4.4.2 Vector Autoregression, One Lag Specification

The results of the VAR-based Granger causality tests, incorporating a one-day lag, are presented in Table 10.

Table 10: Vector Autoregression Granger Causality Tests, One Lag Specification

Null Hypothesis	Chi-squared	*p*-value	Result
Sentiment does not Granger-cause Returns	0.277	.5987	Fail to Reject
Returns do not Granger-cause Sentiment	0.011	.9176	Fail to Reject

The one-lag VAR specification provides no evidence of Granger causality in either direction. The chi-squared statistic for the null hypothesis that sentiment does not Granger-cause returns ($\chi^2 = 0.277$, $p = 0.599$) indicates failure to reject the null, meaning lagged sentiment does not improve prediction of abnormal returns when using a one-day lag. Likewise, the test for reverse causality yields an even smaller chi-squared statistic ($\chi^2 = 0.011$, $p = 0.918$), suggesting no meaningful predictive role of lagged returns for subsequent sentiment.

These results mirror the panel-based Granger causality findings for the one-lag specification and reflect a key aspect of the aggregated VAR approach: with firm-level heterogeneity removed and the common daily series averaged across cases, the dynamics captured at a one-day horizon appear limited. In essence, when restricted to a single lag, the VAR concludes that neither sentiment nor returns systematically anticipate each other.

4.4.3 Vector Autoregression, Two Lag Specification

Given that short-horizon dynamics may not be fully captured by a single lag (particularly in contexts where information diffusion or investor processing occurs over more than one trading day) I estimate an alternative VAR with two lags. The two-lag model allows the system to capture potentially slower adjustment in how sentiment shocks propagate into returns and provides a closer analogue to the two-lag panel Granger causality results presented earlier. Table 11 reports the associated Granger causality tests from this VAR(2) specification.

Table 11: Vector Autoregression Granger Causality Tests, Two Lag Specification

Null Hypothesis	Chi-squared	*p*-value	Result
Sentiment does not Granger-cause Returns	4.879	.0872	Reject*
Returns do not Granger-cause Sentiment	0.634	.7284	Fail to Reject

The output in Table 11 confirms the earlier findings that sentiment Granger-causes returns with marginal significance ($\chi^2 = 4.88$, $p = 0.087$), while returns do not Granger-cause sentiment ($\chi^2 = 0.63$, $p = 0.728$). The p-value of 0.087 on sentiment Granger-causing returns is slightly above the conventional 0.05 threshold, but still provides meaningful evidence given the inherent difficulty of predicting stock returns and the small sample size of five firms.

4.5 Summary of Results

Across all empirical specifications, there are several consistent patterns. First, contemporaneous sentiment shows no meaningful association with same-day abnormal returns, suggesting that market responses to AI feature announcements do not occur instantaneously. Instead, the primary predictive relationship appears with lagged sentiment, and only under specifications that allow for slower adjustment dynamics. Both the two-lag panel Granger causality tests and the two-lag VAR indicate that sentiment contains incremental information about next-day abnormal returns, whereas one-lag models show no such effect. This pattern implies that markets incorporate consumer reactions with a short delay, consistent with partial underreaction rather than immediate price adjustment.

Second, reverse causality receives little support. Across all lag structures and in both panel and VAR frameworks, lagged returns fail to predict sentiment. This asymmetry strengthens the interpretation that sentiment is forward-looking information rather than a reaction to stock performance.

Third, the impulse response analysis shows that sentiment shocks generate small, short-lived movements in returns that dissipate within a few days, consistent with rapid correction after an initial underreaction.

Finally, heterogeneity across firms suggests that the sentiment–return relationship is moderated by product context and user expectations, with Adobe showing stronger alignment between sentiment and returns and other firms exhibiting weaker or even negative associations.

Collectively, the results indicate that consumer sentiment surrounding AI feature announcements does contain predictive information for short-term stock performance, but the effect is modest, asymmetric, and highly dependent on lag structure. These findings reinforce the view that markets process consumer reactions with a brief delay and that sentiment matters most in the immediate aftermath of highly visible product announcements.

5. Conclusion

This study finds that consumer sentiment surrounding AI feature announcements predicts abnormal stock returns, but only under lag structures that allow for slower adjustment dynamics. Under the two-lag Granger causality specification, a one-unit increase in Reddit sentiment is associated with approximately 2.2 percentage points higher abnormal returns the following day. Critically, this relationship flows in only one direction: sentiment predicts returns, but returns do not predict sentiment, and this asymmetry appears consistently across both one- and two-lag specifications. This suggests that consumer reactions on Reddit contain information about firm performance that markets take time to incorporate into prices. The result that predictability emerges only in the two-lag specification indicates that markets incorporate sentiment with a short delay rather than instantaneously.

The timing of these findings is especially relevant. In October 2025, the Bank of England warned that equity valuations for AI-focused companies "appear stretched," and the IMF's managing director noted that valuations are approaching levels last seen during the dot-com bubble (Bank of England 2025; Georgieva 2025). If markets were efficiently pricing AI features based on fundamental value creation, all available information would immediately be reflected in stock prices and as such consumer sentiment should have no predictive power. The conclusion that publicly observable Reddit discussions help predict next-day returns suggests that markets are not immediately incorporating consumer reactions into valuations. This delayed response could reflect the difficulty of assessing AI feature value in real time, or it could indicate speculative momentum that regulators have flagged as a concern.

The results also reveal substantial heterogeneity across firms. Adobe shows a positive sentiment effect, while Microsoft, Google, and Snapchat show negative relationships. This variation is reasonable given the different contexts of each announcement. Adobe's Firefly was a targeted tool for creative professionals, a clear use case with obvious value. Microsoft's Copilot and Google's Gemini, by contrast, integrate AI into core products in ways that users cannot opt out of. Spotify's AI DJ falls somewhere in between: generally well-received but with uncertain revenue implications. Snapchat's My AI displays the most extreme post-announcement price response, with CARs rising sharply despite volatile sentiment, consistent with short-run speculative dynamics. These differences suggest that the market impact of consumer sentiment depends on how much genuine uncertainty exists about product reception and whether the AI feature addresses real user needs or appears as an unwanted addition.

The VAR results mirror these patterns directionally but are weaker in statistical strength due to firm averaging and limited sample size. The nature of sentiment effects seems short lived, dissipating within roughly three days. This indicates that markets quickly correct. The impulse response function shows an initial positive response to sentiment shocks followed by rapid stabilization. This pattern could reflect initial underreaction followed by correction, or brief overreaction followed by reversion to fundamentals. Either interpretation suggests that sentiment's influence is temporary, which is somewhat encouraging with regards to efficiency. However, if these short-term fluctuations are driven by consumer reactions rather than fundamental analysis, they nevertheless contribute to the type of volatility that concerns financial regulators.

Prior research has shown that Reddit communities can influence market outcomes, most notably during the GameStop trading episode. This study extends that finding to intimate that Reddit sentiment also matters for established technology firms when they announce AI features. The mechanism is likely different. This is not a matter of retail traders coordinating to drive prices, but rather consumer feedback providing a signal about product-market fit that professional investors observe and act on. The determination that this signal takes a day to be priced in suggests that institutional investors either do not monitor Reddit in real time (naturally) or need time to aggregate dispersed reactions before adjusting positions.

5.1 Limitations and Future Research

Several limitations should be noted. The sample includes only five firms over short event windows, limiting generalizability. Reddit users skew younger and more tech-savvy than the general population, so their reactions may not represent all consumers. The study examines only short-term price movements and cannot speak to whether sentiment predicts long-term firm performance or product success. The sentiment analysis tools, while validated for social media text, may miss sarcasm, nuance, or context-dependent meaning. Finally, the benchmark used for abnormal returns (the NASDAQ) may not fully capture technology sector dynamics.

Future research could expand the sample to include more AI announcements and firms, examine longer-term outcomes, and compare Reddit sentiment to other sources of consumer opinion. A gauge of overall consumer outlooks and sentiment (like that produced by the University of Michigan) could provide a metric by which announcement-specific consumer sentiment could be compared (University of Michigan Survey of Consumers 2025). It could also prove valuable to investigate the mechanisms more directly: which types of investors respond to sentiment signals, and how quickly do they act? As AI features proliferate, it will be interesting to see whether sentiment's predictive power diminishes as investors gain experience evaluating consumer reactions.

5.2 Practical Implications

For investors, these findings suggest that monitoring consumer sentiment could provide a modest edge in anticipating short-term price movements. However, the small magnitude of effects and the one-day lag limit practical trading opportunities, especially after transaction costs. As more investors adopt sentiment-based strategies, any predictive power will likely erode. The results are likely most useful for understanding market dynamics rather than for generating trading profits.

For firms, the findings emphasize the importance of managing consumer reactions when launching AI features. Even temporary stock price effects have consequence for market perception and executive compensation tied to stock performance. The heterogeneity across firms suggests that companies should carefully evaluate whether their AI integrations address genuine user needs or risk backlash. Adobe's targeted approach appears to have resonated with consumers, while more expansive AI integrations faced more skepticism. Understanding these differences could help firms avoid negative sentiment that, even if short-lived, creates volatility and undermines confidence.

5.3 Final Remarks

Broadly, this study contributes to ongoing debates about AI sector valuations and market efficiency. The fact that consumer sentiment, a form of publicly available information, predicts returns under certain lag structures challenges the notion challenges the notion that markets instantly incorporate all available information into prices (Efficient Markets Hypothesis). At the same time, the transitory nature of effects and the moderate magnitude suggest that markets are not wildly inefficient. The predictive power of sentiment may simply reflect the inherent difficulty of valuing AI innovations in real time, a challenge that becomes even more acute as regulators warn about stretched valuations and bubble risks.

The rapid integration of AI into consumer products has cultivated an environment where firms race to demonstrate AI capabilities while consumers and investors struggle to assess their value. In this context, consumer sentiment serves as a rough but imperfect signal. That markets rely on this signal reveals something about the uncertainty surrounding AI valuations. Whether this represents a temporary adjustment period as markets learn to value AI features or a more fundamental challenge in assessing speculative technology remains an open question. But given the scale of capital flowing into AI and the warnings from financial regulators, understanding how consumer reactions shape market outcomes has never been more important.

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